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4 February 1988
SUPERSEDING
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MILITARY SPECIFICATION

SEMITRAILER, LOWBED: TACTICAL, 12- AND 25-TON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 12- and 25-ton lowbed semitrailers intended for use over all types of roads as tactical vehicles during military operations (see 6.1).

1.2 Classification. Semitrailers shall be classified by model, as follows (see 6.2):

Model	Description
M172A1	Semitrailer, Lowbed, 25-Ton, 4-Wheel.
M270A1	Semitrailer, Lowbed, 12-Ton, 4-Wheel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS
MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-12459	- Tire, Pneumatic: for Military Ground Vehicles.
MIL-R-46164	- Rustproofing for Military Vehicles and Trailers.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Simple Component, Chemical Agent Resistant.
MIL-V-62038	- Vehicles, Wheeled: Preparation for Shipment and Storage of.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-209	- Slinging and Tie Down Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

8358925	- Semitrailer, Wrecker, 12-Ton, 4-Wheel, M270A1.
8736073	- Semitrailer, Lowbed, 25-Ton, 4-Wheel, M172A1.
12355846	- Treatment and Paint Specifications for Tactical Trailers.

(Copies of specifications, standards, and drawings required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish semitrailers, of specified model, which shall be subjected to first article inspection (see 4.3). First article inspection samples, properly marked with identifying information shall be representative of the semitrailers to be furnished to the Government. All subsequent semitrailers delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Materials shall be free of all defects and imperfections that might affect the serviceability and appearance of the finished product (see 4.8.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4.1).

3.3 Design and construction. Completed vehicles shall conform to this specification and the applicable drawing listed below (see 4.8.1 and 4.8.2).

Model	Drawing number
M172A1	8736073
M270A1	8358925

3.3.1 Qualified products list (QPL). The contractor shall be responsible for using parts and assemblies necessary in vehicle construction that are listed as qualified products on the QPL and those approved for inclusion on the QPL. The contractor's manufacturing inspection records shall specifically list all QPL items by part or drawing number(s), name of supplier(s), and number and date of the QPL. When assemblies and parts are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document with the name of suppliers(s) (see 4.1.4 and 4.8.1).

3.3.2 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings (see 4.8.1).

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3.3.3 Seals. When the semitrailer is fording or operating in mud, sand or snow, the bearing seals shall prevent entrance of foreign matter into the bearings which are exposed to contamination during these operations. Water contamination of wheel bearing lubricant, after fording operations, shall be not greater than 2 percent (%). All bearing seals shall restrict the leakage of lubricants from bearings (see 4.8.2).

3.3.4 Electrical.

3.3.4.1 Electrical circuits. Electrical circuits shall maintain continuity from end to end, without evidence of internal or external shorts, during all semitrailer operating conditions. All electrical cables and wiring shall be anchored in such a manner to preclude any damage to cables and wiring during all semitrailer operating conditions (see 4.8.2).

3.3.4.2 Lights. All lights specified on applicable drawings shall operate throughout all semitrailer operating conditions (see 4.8.2).

3.3.5 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction, throughout all ranges of operation, under all semitrailer operating conditions (see 4.8.2).

3.3.6 Adjustment mechanisms. All adjustment mechanisms shall function properly, and maintain adjustment settings during all semitrailer operating conditions (see 4.8.2).

3.3.7 Welding. Welding shall meet the requirements specified on the applicable drawing (see 4.8.1 and 4.8.2).

3.3.7.1 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the acquisition activity (see 4.8.1 and 6.2).

3.3.8 Tie down devices. Unless otherwise specified (see 6.2), semitrailers shall be provided with tie down devices, of sufficient number and structural strength as to be adequate for transport by rail, sea and air (see 4.8.2).

3.3.9 Air and hydraulic lines, and fittings. Provisions shall be made to assure internal cleanliness of air and hydraulic lines and fittings, prior to initial use of semitrailer. All lines and connections shall be free of leaks and shall be anchored in such manner as to preclude damage during all semitrailer operating conditions (see 4.8.2).

3.3.10 Tires. The semitrailer shall be furnished with tube type tires conforming to class HW, style A of MIL-T-12459 (see 4.8.2).

3.3.11 Lifting attachments. The semitrailer shall be provided with lifting attachments that are structurally adequate to lift the semitrailer. Lifting attachments shall be capable of lifting the semitrailer when loaded with maximum payload, as specified below. Lift slings, with spreader bar, shall conform to MIL-STD-209 (see 4.8.2 and 4.8.3).

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Model	Maximum payload [pounds (lbs)]
M172A1	50,000
M270A1	24,000

3.3.12 Landing legs. Landing legs shall suppport the fully loaded semitrailer and withstand, without damage, the strains imposed when coupling and uncoupling the semitrailer from the towing vehicle and raising and lowering the loaded semitrailer. The landing gear shall operate freely when vehicle is loaded or unloaded and shall operate without interference (see 4.8.2 and 4.8.4).

3.4 Performance. The semitrailer shall meet the performance requirements of 3.4.1 through 3.4.8 of this specification when equipped as specified herein and as depicted on the drawings, serviced as specified in the applicable lubrication drawing, loaded as specified in table I and coupled to the towing vehicle. The semitrailer, serviced and equipped for existing climatic conditions, shall operate as specified without special equipment (see 4.8.5).

TABLE I. Maximum weights, loads and speeds.

Semi-trailer	Vehicle weight (lbs)	Payload (lbs)	Speeds (mph)	
			Highway	Cross-country
M172A1	16,500	50,000	30	10
M270A1	17,500	24,000	20	10

3.4.1 Payload, towing speeds and trailing ability.

3.4.1.1 Highway operation. The semitrailer shall provide satisfactory operation over smooth, hard-surfaced roads, when loaded with payload and towed at the sustained speed specified in table I. When towed at specified speed in a straight line, semitrailer shall track the towing vehicle without shifting or swerving more than 3 inches from side to side of the path of the towing vehicle (see 4.8.5.1).

3.4.1.2 Cross-country operation. The semitrailer, loaded with rated payload uniformly distributed over the load area, shall be towed over unimproved roads, open fields, rolling hills and cross-country terrain at the sustained speed specified in table I without damage to the semitrailer or towing vehicle. The payload shall be distributed so that the proper center of gravity is established (see 4.8.5.2).

3.4.2 Grade and slope operation. When towed over a dry, hard surface free of loose material, the semitrailer shall meet the requirements of 3.4.2.1 and 3.4.2.2.

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3.4.2.1 Longitudinal inclines. The semitrailer shall follow the towing vehicle without weaving to an extent affecting the controllability of the vehicle combination, when ascending and descending longitudinal inclines having a 20% grade (see 4.8.5.3).

3.4.2.2 Side slopes. The semitrailer shall follow the towing vehicle without slipping or upsetting, when operating on left and right side slopes having a 20% grade (see 4.8.5.4).

3.4.3 Braking.

3.4.3.1 Service brakes. The service brakes of the semitrailer and towing vehicle shall decelerate and stop the vehicle combination, with rated payload, on dry, smooth, level, hard-surfaced roads, free of loose materials, as follows (see 4.8.2 and 4.8.5.5):

a. M172A1 - The stopping distance shall be not more than 40 feet (ft) from a speed of 20 miles per hour (mph).

b. M270A1 - The stopping distance shall be not more than 50 ft from a speed of 20 mph.

3.4.3.2 Automatic brake actuator. The actuator shall automatically apply the semitrailer service brakes upon breakaway from towing vehicle. The actuator shall maintain the application of brakes holding the trailer stationary on a 30% grade under all loads specified herein for not less than 15 minutes (see 4.8.2 and 4.8.5.6).

3.4.4 Turning ability. The semitrailer-towing vehicle combination shall be capable of making complete turns to the right or left, with the semitrailer assuming a 90 degree angle without cramping, side-slipping or damage to either the towing vehicle or the semitrailer (see 4.8.5.7).

3.4.5 Fording ability. The semitrailer shall be capable of being towed in fresh or salt water to a depth of not less than 5 ft for periods up to 15 minutes, without impairing performance of the semitrailer or its components. Fording shall be accomplished without use of special kits, and with a minimum of preparation or servicing before and after the fording operation (see 4.8.5.8 and 6.5).

3.4.6 Environmental. The semitrailer shall be capable of being towed under extreme conditions of weather and altitude in ambient temperatures ranging from minus 65 degrees Fahrenheit (°F) to plus 125°F. The semitrailer shall be capable of withstanding storage at minus 80°F without deterioration that cause failure of any component (see 4.8.5.9).

3.4.7 Reliability. The semitrailer shall achieve not less than 6000 mean miles between failure (MMBF) when operated in accordance with the 6000-mile test scenerio specified in 4.3.2.2 (see 4.8.5.10).

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3.4.7.1 Failure. A failure is any malfunction of an end item requiring overhaul, replacement, rebuild, salvage or other corrective action as prescribed by the maintenance manual, which cannot be deferred for the remainder of 6000 miles.

3.4.7.2 Corrective action. Corrective action is not deferrable if malfunction causes, or would cause if not corrected:

a. Inability to commence operation or cessation of operation.
b. Inability to conform to the requirements of the specification.
c Critical or catastrophic hazard as defined in MIL-STD-882.

3.4.7.3 Malfuntion. Any malfunction which the operator or crew can remedy will not be considered a failure provided that repair is authorized or prescribed, as an operator function, and can be accomplished within 1 hour using only controls, tools, or spare parts incorporated in, or carried with, the end item, or its towing vehicle.

3.4.8 Maintainability. Total scheduled maintenance excluding driver and crew checks and services shall not exceed 8 manhours during the first 6000 miles of operation in accordance with the test scenerio specified in 4.3.2.2. This equates to a maintenance ratio (MR) of 0.027 at 20 operational miles equivalent to 1 operational hour. The scheduled maintenance interval shall be 6 months or 6000 miles, whichever comes first (see 4.8.5.11).

3.5 Painting. Unless otherwise specified (see 6.2), the exterior and interior of the semitrailer, component assemblies and parts which require painting shall be cleaned, treated and CARC painted in accordance with MIL-STD-193 and Drawing 12355846 (see 4.8.2).

3.5.1 Protective coating. When specified (see 6.2), all parts of sheet and structural metal surfaces shall be rustproofed in accordance with MIL-R-46164 (see 4.8.2).

3.6 Marking. Semitrailer marking shall be in accordance with MIL-STD-642. Markings on Army semitrailers shall be in black, lusterless paint having the equivalent infrared reflectance of MIL-C-46168 or MIL-C-53039. Army semitrailer registration numbers shall be placed on both sides and the characters shall optionally be 4 or 3 inches in height (see 4.8.2).

3.6.1 Components. All parts, assemblies and subassemblies requiring identification shall be identified in accordance with MIL-STD-130 (see 4.8.2).

3.6.2 Identification plates. Identification plates shall conform to MIL-P-514. Any special information required by the using service shall be as specified by the acquisition activity (see 4.8.2 and 6.2).

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3.7 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government, the contractor shall adjust and service each semitrailer for immediate operational use, including at least the following: Adjust braking system, check electrical system, inflate all tires, and completely lubricate the semitrailer and all running gear with grades of lubricants recommended for the ambient temperature at the delivery point (see 4.8.2).

3.8 Safety. No condition shall exist which may create a safety hazard to operating or maintenance personnel. All exposed parts which are energized electrically shall be located, insulated, fully enclosed or guarded so as to prevent hazards to operating personnel or equipment malfunction. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions (see 4.9).

3.9 Workmanship. Workmanship shall be of such quality as to assure that the semitrailer and its components are free of defects that compromise, limit or reduce the capability of the semitrailer in the performance of its intended use. In addition, the vehicle shall be free of defects which affects its appearance (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operation to determine that practices, methods, and procedures of the written quality assurance system plan are being properly applied, and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedure, or instances of poor practices which might have an adverse effect upon quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until the conformance of the product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. When a part or component is specified to conform to a specification having a QPL, the contractor shall make available to the Government documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order and quantity.

4.1.5 Quality assurance provisions. In the conduct of inspection, the contractor shall adhere to Quality Assurance Provisions (QAP) and General Quality Assurance Provision (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.1.6 Certification. Where certification (see 4.8.1) is required to verify material or component conformance to a specification, the contractor shall maintain such certification along with documented test results and performance and analytical data, as applicable.

4.2 Classification of inspection:

a. First article inspection (see 4.3).
   1. First production vehicle inspection (FPVI) (see 4.3.1).
   2. Initial production test (IPT) (see 4.3.2).

b. Quality conformance inspections (QCI) (see 4.4).
c. Control tests (see 4.5).
d. Comparison tests (see 4.6).

4.3 First article inspection. First article inspection shall be performed on the first production vehicle and on 2 additional vehicles, randomly selected from the first month's production, or the first 20 produced which shall be designated as the initial production vehicles. Approval of the first article samples shall not relieve the contractor of the obligation to supply vehicle(s) that are fully representative of those inspected as a first article sample. Any change or deviation of production vehicles from the first article sample shall be subject to the approval of the contracting officer.

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4.3.1 FPVI.

4.3.1.1 In-process inspection. The first production vehicle shall be inspected during fabrication to determine conformance of materials and workmanship to specified requirements. These inspections shall preceed application of primer and paint. Processing and welding procedures, the quality system, inspection records, calibration procedures, radiographic procedures and welder certifications shall be reviewed and evaluated.

4.3.1.2 Completed vehicle inspections.

4.3.1.2.1 Contractor inspection. The contractor shall subject the vehicle to a road test and the FPVI specified in table II. The road test shall be performed in accordance with 4.5.1.

4.3.1.2.2 Preliminary inspection. The responsible Government inspection element shall conduct the FPVI specified in table II to the extent that acceptability of the vehicle is verified.

4.3.1.2.3 Provisional inspection. The Government acquisition activity shall perform the FPVI specified in table II to the degree required to justify approval of the vehicle.

TABLE II. Classification of inspections.

Title	Requirements	Inspection	First article 1/ FPVI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Materials and construction	3.2 thru 3.3.2, 3.3.7 and 3.3.7.1	4.8.1	X	X			
Defects (see table IV)	3.3, 3.3.3 thru 3.3.7, 3.3.8 thru 3.3.12, 3.4.3.1, 3.4.3.2, 3.5 thru 3.7 and 3.9	4.8.2	X	X	X		
Lifting attachments	3.3.11	4.8.3	X	X	X		
Landing legs	3.3.12	4.8.4	X	X	X		
Highway operation	3.4.1.1	4.8.5.1	X	X	X	X	
Cross-country operation	3.4.1.2	4.8.5.2	X	X			X
Longitudinal inclines	3.4.2.1	4.8.5.3	X	X		X	

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TABLE II. Classification of inspections - Continued.

Title	Requirements	Inspection	First article 1/ FPVI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Side slopes	3.4.2.2	4.8.5.4	X	X		X	X
Service brakes	3.4.3.1	4.8.5.5	X	X	X	X	
Automatic brake actuator	3.4.3.2	4.8.5.6	X	X	X	X	
Turning ability	3.4.4	4.8.5.7	X	X	X		
Fording ability	3.4.5	4.8.5.8		X			X
Environmental	3.4.6	4.8.5.9		X			X
Reliability	3.4.7	4.8.5.10		X			
Maintainability	3.4.8	4.8.5.11		X			
Safety	3.8	4.9		X			

1/ At place of manufacture
2/ At Government proving ground

4.3.1.2.4 Vehicle disposition. After completion of inspection, the first production vehicle shall remain at the manufacturing facility, as a production sample, and be the last vehicle of its type shipped on the contract. The vehicle may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the vehicle during this period in accordance with applicable documents for care and preservation while in storage. All configuration changes taking place after the first production vehicle inspection, shall be made to the first production vehicle so that the vehicle will be representative of the current configuration throughout the life of the contract. No configuration changes may be implemented on subsequent production vehicles after Government conditional acceptance of the first production vehicle (applicable vehicle model) without written authorization from the Government.

4.3.2 IPT.

4.3.2.1 Contractor inspection. The contractor shall inspect the IPT vehicles in accordance with 4.4.

4.3.2.2 Government inspection. The Government shall perform the IPT specified in table II. The two vehicles shall be representative of production types. The selected vehicles shall be examined for defects. Subsequent to examination, the vehicles shall be subjected to a 6000-mile test in accordance with table III, at a site(s) selected by the Government, and the tests specified in table II. The contractor shall expeditiously furnish repair parts, as required, to adequately support the tests. In the

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event the production contract delineates test mileage less than 6000 miles, the mileage mix for the various types of courses in table III shall be proportional to the miles shown in the mileage and speed column. The vehicles shall be tested with actual or simulated payload. Unless otherwise specified (see 6.2), the contractor shall refurbish vehicles after test completion.

TABLE III. 6000-mile test.

Course	Mileage and speed 1/	Payload 1/
Hard-surfaced roads	1200 miles at speeds up to maximum	Highway
Level cross-country roads	3000 miles at speeds up to maximum	Cross-country
Hilly cross-country roads (40% on gravel roads)	1500 miles at speeds up to maximum	Cross-country
Belgium block roads	300 miles at speeds applicable to semitrailer characteristics	Cross-country (Run the first and last 50 miles without payload)

1/ See table I.

4.3.3 Deficiencies. Unresolved failures or deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the IPT shall be evidence that all vehicles already accepted prior to completion of the IPT are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all vehicles shall be corrected by the contractor.

4.4 QCI. QCI shall be performed in the following sequence and utilizing an approved Final Inspection Record (FIR).

a. Examination for the defects specified in table II and classified in table IV.
b. A 5-mile road test in accordance with 4.4.5.
c. QCI specified in table II.

4.4.1 One hundred percent. Each vehicle shall be subjected to the QCI specified in 4.4.

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4.4.2 Waiver of 100% QCI. When authorized by the Government contracting authority, 100% QCI shall be waived. Thereafter, samples shall be selected in accordance with 4.4.3 and shall be subjected to the QCI specified in 4.4. For examination purposes, samples shall be inspected to determine conformance to the following acceptable quality level (AQL).

Classification	AQL
Critical	None
Major	25 defects/100 units
Minor	150 defects/100 units

4.4.3 Sampling.

4.4.3.1 Lot formation. An inspection lot shall consist of all the semitrailers of one model produced from an identifiable production period (one day's production) submitted at one time for acceptance.

4.4.3.2 Sampling for inspection. When sampling is authorized for QCI (see 4.4.2), samples shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall examine at least 20 semitrailers of each model consecutively produced. The examination shall establish that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQL (see table IV).

4.4.3.3 Quality conformance examination by sampling. Each vehicle selected by sampling for final inspection shall be inspected by the contractor utilizing a contractor prepared and Government approved FIR to verify conformance of produced vehicles to the requirements of this specification (see table II). Examinations shall be performed using the classification of defects and the AQL specified in table IV.

4.4.3.4 Quality conformance testing by sampling. Each vehicle system produced, whether subjected to a final inspection by sampling (see 4.4.3.2) or not, shall be operated for a distance of not less than 5 miles by the contractor as depicted in 4.4.5. Vehicles shall be completely assembled and serviced prior to road test. Performance of vehicles shall be demonstrated with minimal payload at the place of manufacture on a smooth, relatively level, hard-surfaced road.

4.4.4 Defects. The following rules shall be applied to the counting of defects (see table IV and 6.3).

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TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	25 defects/100 units	
101	Frame and gooseneck - improper components, improperly welded, improperly riveted (see 3.3 and 3.3.7).	Visual
102	Axles - improperly assembled, improperly installed, improperly welded (see 3.3 and 3.3.7).	Visual and functional
103	Suspension system - damage, improper ground clearance (see 3.3).	Visual and functional
104	Fifth wheel plate and kingpin - improper mounting or assembly, improperly welded (see 3.3 and 3.3.7).	Visual and functional
105	Electrical system, cables, wiring, ground straps - malfunction, improper lighting sequence, improper components, improper operation, inoperative, damage (see 3.3.4).	Visual, functional and SIE 1/
106	Controls, electrical, mechanical, hydraulic - malfunction, inoperative (see 3.3.5).	Visual and functional
107	Adjustment mechanism - defective, improper function (see 3.3.6).	Visual and functional
108	Wheels and tires - malfunction, tire damage (see 3.3.10).	Visual and functional
109	Landing legs - malfunction (see 3.3.12).	Visual and functional
110	Service brakes - inoperative, malfunction, component damage, nonconformance, leaking (see 3.4.3.1).	Visual and functional
111	Paint - improper paint, improper compounding or curing (see 3.5).	Visual and functional
112	Faulty workmanship affecting performance (see 3.9).	Visual
Minor	AQL 150 defects/100 units	
201	Record forms - missing, improper information (see 3.3).	Visual
202	Platform, stake pockets, lifting eyes, lashing rings, stowage compartments and tool boxes - improper (see 3.3).	Visual
203	Suspension system - improper assembly or installation (see 3.3).	Visual
204	Seals - leaking, defective, damage (see 3.3.3).	Visual

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TABLE IV. Classification of defects - Continued.

Category	Defect	Method of examination
205	Electrical system - improper assembly or installation, improperly coded (see 3.3.4).	Visual
206	Tiedowns, lashing hooks - missing, improperly located (see 3.3.8).	Visual
207	Air and hydraulic lines, and fittings - improper installation or operation (see 3.3.9).	Visual
208	Wheels and tires - improper size, type or mounting (see 3.3.10).	Visual and SIE
209	Lifting attachments - missing, improperly located, improper welding (see 3.3.7 and 3.3.11).	Visual
210	Landing legs - improper assembly or installation (see 3.3.12).	Visual
211	Brake system - improper assembly, installation or adjustment (see 3.4.3).	Visual and functional
212	Paint - improper application or color (see 3.5).	Visual
213	Protective coating - improper application, improper materials, improper coverage (see 3.5.4).	Visual
214	Marking, identification - missing, improper size, improper installation (see 3.6).	Visual
215	Lube fittings - missing, improper installation, improper operation (see 3.3).	Visual
216	Lubrication - missing, improper grade or application (see 3.7).	Visual
217	Hydraulic fluid - missing, improper grade (see 3.7).	Visual
218	Workmanship - improper, affecting appearance (see 3.9).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.4.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but are considered departures from good workmanship, shall be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots, or 10 lots or more within a 30-day period, shall be added to the minor defects classification with no increase in AQLs (see 6.3).

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4.4.4.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect shall be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

4.4.4.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

4.4.5 Five-mile road test. The semitrailer shall be completely assembled and serviced and shall be towed for a distance of not less then 5 miles without payload. The performance of the semitrailers shall be demonstrated on relatively smooth, hard-surfaced roads.

4.5 Control tests. Control tests shall be conducted on 1 semitrailer per month or 1 out of each 100 semitrailers submitted but not more than 2 semitrailers in any given 30-day period. These semitrailers shall be subjected to the 50-mile road test specified in 4.5.1 and the control tests specified in table II.

4.5.1 Fifty-mile test. Each semitrailer shall be loaded with full or simulated payload and towed for a distance of not less than 50 miles. The test course shall be a relatively smooth, hard-surfaced road.

4.6 Comparison tests. The Government may randomly select semitrailers at any time during the production contract period and subject these semitrailers to the 6000-mile test in accordance with table III and the comparison tests specified in table II. Tests shall be conducted by the Government at a site it shall select. Semitrailers selected for comparison tests shall not have been previously tested according to 4.5, but shall have passed inspection according to 4.4.

4.7 Failure. Failure of any semitrailers to pass any of the specified inspections or failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3.2, 3.3.7 and 3.3.7.1 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.8.2 Defects. Conformance to 3.3, 3.3.3 through 3.3.7, 3.3.8 through 3.3.12, 3.4.3.1, 3.4.3.2, 3.5 through 3.7 and 3.9 shall be determined by examination for the defects listed in table IV. Examination shall be visual, functional, tactile or by measurement with standard inspection equipment.

4.8.3 Lifting attachments. To determine conformance to 3.3.11, the semitrailer, loaded as specified, shall be lifted not less than 1 ft and held for 15 minutes prior to road tests and examined for damage to the semitrailer and lifting devices.

4.8.4 Landing legs. To determine conformance to 3.3.12, the semitrailer shall be loaded with an actual rated payload or with a simulated payload distributed to assure that the landing legs are supporting the intended load. The landing legs shall then be examined for meeting functional requirements and ease of operation.

4.8.5 Performance. To determine conformance to 3.4, the semitrailer shall pass the tests of 4.8.5.1 through 4.8.5.11. Prior to these tests, the vehicle shall be equipped, serviced, loaded and coupled to the towing vehicle as specified herein.

4.8.5.1 Highway operation. To determine conformance to 3.4.1.1, the semitrailer shall be operated as specified on paved and gravel roads and observed for towing and performance requirements.

4.8.5.2 Cross-country operation. To determine conformance to 3.4.1.2, the semitrailer shall be operated over cross-country terrain as specified without damage to the semitrailer or the towing vehicle.

4.8.5.3 Longitudinal inclines. To determine conformance to 3.4.2.1, the semitrailer shall be towed by the towing vehicle as specified. During the operation, the semitrailer shall be observed for its performance. Semitrailer slip or upset shall be considered failure.

4.8.5.4 Side slopes. To determine conformance to 3.4.2.2, the semitrailer shall be towed by the towing vehicle as specified. During the operation, the semitrailer shall be observed for its performance. Semitrailer slip or upset shall be considered failure.

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4.8.5.5 Service brakes. To determine conformance to 3.4.3.1, the brakes of the towing vehicle and semitrailer, with rated payload, shall be applied simultaneously while the combination is traveling forward at 20 mph. Stopping distance in excess of 40 ft for M172A1 semitrailer and 50 ft for M270A1 semitrailer shall constitute failure of this test.

4.8.5.6 Automatic brake actuator. To determine conformance to 3.4.3.2, the loaded trailer shall be separated from the towing vehicle while both are moving. The automatic device shall actuate the brakes which shall stop and hold the trailer stationary for 15 minutes. Failure to stop, or beginning to roll in less than 15 minutes after stop, shall constitute failure of this test.

4.8.5.7 Turning ability. To determine conformance to 3.4.4, the towing vehicle shall be coupled to the semitrailer, and the combination shall be driven making complete turns to the right and left with the semitrailer assuming a 90 degree angle without interference between the towing vehicle and the semitrailer. Cramping, side-slipping or damage to either the towing vehicle or the semitrailer shall constitute failure of this test.

4.8.5.8 Fording ability. To determine conformance to 3.4.5, the semitrailer shall be placed in or towed through water at the depth and for the duration specified. All the sealed items and bearings shall be examined for water contamination. Contamination for sealed bearings shall not be in excess of that specified in 3.3.3.

4.8.5.9 Environmental. To determine conformance to 3.4.6, the semitrailer shall be operated and stored at the temperatures specified. The semitrailer shall operate as specified and evidence no damage as a result of such operation or storage.

4.8.5.10 Reliability. To determine conformance to 3.4.7, a minimum acceptable value (MAV) of 6000 miles shall be demonstrated during testing specified in 4.3.2.2. The point estimate MAV shall be computed using the cumulative number of chargeable failures of all vehicles.

4.8.5.11 Maintainability. To determine conformance to 3.4.8, a MR of less than, or equal to, that specified in 3.4.8 shall be demonstrated during testing specified in 4.3.2.2. The MR shall be calculated using the total cumulative maintenance manhours with the exceptions of maintenance induced errors and driver/crew checks and services.

4.9 Safety. To determine conformance to 3.8, the semitrailer and its components shall be examined.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with MIL-V-62038 and the applicable packaging requirements specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. The semitrailers covered by this specification are intended for transporting equipment and general cargo over all types of roads during military operations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Model of the semitrailer required (see 1.2).
c. If first article inspection is not required (see 3.1).
d. If welding repairs are authorized and extent allowed (see 3.3.7.1).
e. If tie down devices are not required (see 3.3.8).
f. If painting requirements shall be other than as specified (see 3.5).
g. If protective coating (rustproofing) is required (see 3.5.1).
h. Special identification marking required by the user on identification plates (see 3.6.2).
i. If responsibility for inspection shall be other than as specified (see 4.1).
j. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
k. If vehicles shall not be refurbished after IPT (see 4.3.2.2).
l. Selection of applicable level and packaging requirements (see 5.1).

6.3 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each vehicle is performed by the contractor for the listed characteristics, and for any departures from good workmanship. The Government inspector shall assure that all defects encountered during inspection are enumerated on the deficiency sheet for the vehicle. Defects noted on a deficiency sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the defects in accordance with the classification of defects (see table IV) and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring defects (see 4.4.4.2 and 4.4.4.3).

6.4 Definitions.

6.4.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.2).

6.4.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in

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domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 International standardization agreement. Certain provisions of this specification (see 3.4.5) are the subject of international standardization agreement NATO STANAG 2805. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.6 Subject term (key word) listing.
Lowbed semitrailer, tactical
Semitrailer, lowbed, tactical
Tactical, semitrailer, lowbed

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - MC

Preparing activity:
Army - AT

(Project 2330-0957)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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